MILITARY SPECIFICATION

POWER SUPPLY, MULTIVOLTAGE, DIRECT CURRENT (FOR TORPEDO TEST SET MARK 281 MOD 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers a power supply with five different voltage outputs for use with the Torpedo Test Set Mark 281 Mod 1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods

FSC 6130

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DRAWINGS

Naval Ordnance Systems Command

2639690	Power Supply Schematic Diagram
2639691	Power Supply Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First articles. Unless otherwise specified in the contract or purchase order, three first articles of the power supply shall be manufactured using the methods and procedures proposed for production. The first articles will be tested as specified in section 4 to determine if the proposed production method will yield an item which meets the technical and design requirements of this specification and the contract or purchase order.

3.2 Material. Unless otherwise specified in the contract or purchase order, all materials and purchased parts entering into the final assembly of the power supply shall conform to NAVORD Drawing 2639691.

3.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions shall be in accordance with NAVORD Drawing 2639691 and MIL-STD-454, requirements 1, 3, 15, and 62, as applicable.

3.4 Moisture, fungus, and corrosion. The power supply shall be moisture, fungus, and corrosion resistant in accordance with NAVORD Drawing 2639691 and MIL-STD-454, requirements 4, 16, and 31, as applicable.

3.5 Interchangeability. Interchangeability shall be as specified in MIL-STD-454, requirement 7.

3.6 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein, shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification, resolution, or correction.

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The generator set and cable ends shall be matchmarked with cage identifying

3.7 Operation. The power supply shall operate as specified by NAVORD Drawing 2639690.

3.8 Performance.

3.8.1 Input power. The power supply shall operate with an input power supply of 115 volts alternating current ±5 percent at 60 hertz ±5 percent.

3.8.2 Output power. Five separate output voltages shall be available with the capabilities shown in table I.

Table I

OUTPUT POWER

Voltage			Amperes	Load		Ripple voltage (v)
Nominal	Max.	Min.	Nominal	Min.	Max.	max. at load
-18 dc	-16.3	-19.0	1.0	Open[1]	18.0 ohms	0.1 v(P-P)[2] 36.0 ohms
6 dc	6.9	5.3	0.5	Open[1]	12.0 ohms	0.1 v(P-P)[2] 23.0 ohms
-24 dc	-21.7	-25.0	2.5	Open[1]	10.0 ohms	0.2 v(P-P)[2] 16.6 ohms
-24 dc	-18.0	-32.0	10	Open[1]	2.5 ohms	-[3]
33.5 ac	36.1	30.9	0.3	Open[1]	100 ohms	-

[1]Open load—no load, no connection to output terminals.
[2]P-P—peak-to-peak.
[3]Under a load of 2.5 ohms, the duty cycle shall be no less than 0.1 second per 5-minute period.

3.9 Environmental requirements. The power supply shall withstand the following environmental conditions without damage:

(a) High temperature, operating: 73° Celsius
(b) Low temperature, operating: 0° Celsius
(c) High temperature, storage: 85° Celsius
(d) Low temperature, storage: -62° Celsius
(e) Vibration, operating: MIL-STD-167, type I equipment
(f) Vibration, nonoperating: 3.5 g peak at 5 to 500 hertz
(g) Shock: 40 g for 18 milliseconds
(h) Humidity, nonoperating: 95 percent relative at 56° Celsius
(i) Humidity, operating: 85 percent relative at 56° Celsius
(j) Salt fog atmosphere, nonoperating.

3.10 Weight. The power supply shall not weigh more than 20 pounds.

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3.11 Marking. The power supply assemblies, subassemblies, and parts shall be marked in accordance with NAVORD Drawing 2639691 and MIL-STD-454, requirements 34 and 67, as applicable.

3.12 Workmanship. The workmanship of the power supply shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the power supply shall be classified as follows:

(a) First article inspection
(b) Quality conformance inspection.

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order, three power supplies shall be manufactured using the methods and procedures proposed for production. These power supplies shall be subjected to the first article inspection procedure of 4.4.

4.2.2 Quality conformance inspection. The procedure of 4.5 shall be performed on each power supply submitted for acceptance under contract (100 percent inspection) to assure that production units meet the requirements of this specification and the contract or purchase order.

4.3 Test conditions. Unless otherwise specified herein, the power supply shall be tested under the following ambient conditions:

(a) Temperature: 25° ± 10° Celsius
(b) Humidity: not exceeding 95 percent relative.

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4.3.1 The input power supply shall provide 115 ± 1 volts alternating current at 60 ± 1 hertz and shall be connected as specified in NAVORD Drawing 2639690.

4.3.2 The power supply shall be adjusted to meet the requirements of note 1 in NAVORD Drawing 2639690 prior to the start of testing.

4.4 First article inspection procedure.

4.4.1 First article inspection shall be performed on three first articles in accordance with the test schedule shown in table II.

4.4.2 A failure of any power supply in any test or inspection shall be cause for denial of production approval.

4.5 Quality conformance inspection procedure.

4.5.1 Quality conformance inspection shall be performed on each power supply submitted for acceptance under the contract according to the test schedule shown in table II.

4.5.2 A failure of any power supply in any test or inspection shall be cause for rejection of that power supply.

4.6 Visual and mechanical examination.

4.6.1 The complete power supply shall be weighed on a scale accurate to within ±0.25 pound.

4.6.2 The complete power supply shall be examined visually to determine conformance to the requirements of 3.2, 3.3, 3.4, 3.11, and 3.12.

4.7 Test methods.

4.7.1 Operation test. The power supply shall be subjected to each of the tests called out under note 2 on NAVORD Drawing 2639690.

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Table II

TEST SCHEDULE

Test	Requirement	Method	First article unit			Quality conformance
			A	B	C	
Visual mechanical examination	3.2, 3.3, 3.4, 3.5, 3.10, 3.11, 3.12	4.6	X	X	X	X
Operation	3.7	4.7.1	X	X	X	X
Performance	3.8	4.7.2	X	X	X	
High temperature, operating	3.9	4.7.3.1	X	X		
Low temperature, operating	3.9	4.7.3.2	X	X		
High temperature, storage	3.9	4.7.3.3	X		X	
Low temperature, storage	3.9	4.7.3.4	X		X	
Vibration, operating	3.9	4.7.3.5	X		X	
Vibration, nonoperating	3.9	4.7.3.6	X		X	
Shock	3.9	4.7.3.7	X	X		
Humidity	3.4, 3.9	4.7.3.8	X	X		
Salt fog	3.4, 3.9	4.7.3.9		X		
Fungus	3.4	4.7.3.10		X		

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4.7.2 Performance test.

4.7.2.1 The following tests shall be performed under the test conditions of 4.3, unless otherwise specified.

4.7.2.2 Low input voltage test.

4.7.2.2.1 The input power supply shall provide 109.4 volts alternating current ±0.1 percent at 60 ± 3 hertz and shall be connected as specified in NAVORD Drawing 2639690.

4.7.2.2.2 The five output voltages shall be measured at the load limits specified in table I using a suitable device accurate to within ±0.1 volt.

4.7.2.3 High input voltage test.

4.7.2.3.1 The procedure of 4.7.2.2 shall be repeated except the input power supply shall provide 120.6 volts alternating current ±0.1 percent at 60 ± 3 hertz.

4.7.3 Environmental tests.

4.7.3.1 High temperature, operating. The power supply, less packaging, shall be subjected to a temperature of 71.1° ± 1.7° Celsius for a period of 3 hours minimum. Within 15 minutes after removal from the temperature chamber, the power supply shall be subjected to the procedure of 4.7.1.

4.7.3.2 Low temperature, operating. The power supply, less packaging, shall be subjected to a temperature of -12.5° ± 2.5° Celsius for a period of 6 hours minimum. Within 15 minutes after removal from the temperature chamber, the power supply shall be subjected to the procedure of 4.7.1.

4.7.3.3 High temperature, storage. The power supply, less packaging, shall be subjected to a temperature of 82.5° ± 2.5° Celsius for a period of 24 hours minimum. After returning to room temperature, the power supply shall be subjected to the procedures of 4.6 and 4.7.1.

4.7.3.4 Low temperature, storage. The power supply, less packaging, shall be subjected to a temperature of -58.5° ± 3.5° Celsius for a period of 72 hours minimum. After returning to room temperature, the power supply shall be subjected to the procedures of 4.6 and 4.7.1.

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4.7.3.5 Vibration, operating. The power supply, less packaging, shall be subjected to type I mechanical vibration specified in MIL-STD-167. The power supply shall be energized during this test and shall be subjected to the performance test procedures of 4.7.2, while vibrating in each of its three principal axes. At completion of this test, the power supply shall be examined for loose components or other physical damage.

4.7.3.6 Vibration, nonoperating. The power supply, less packaging, shall be subjected to vibration cycling in accordance with MIL-STD-810, method 514, procedure X, curve AY for a period of 1 hour in each of the three axes. At the completion of this test, the power supply shall be examined for physical damage and shall be subjected to the performance test procedures of 4.7.2.

4.7.3.7 Shock. The power supply, less packaging, shall be subjected to the shock test of MIL-STD-810, method 516, procedure I, figure 516.1-2, amplitude b, duration d. At the completion of this test, the power supply shall be examined for physical damage and shall be subjected to the performance test procedures of 4.7.2.

4.7.3.8 Humidity. The power supply shall be subjected to the humidity tests of MIL-STD-810, method 507, procedure I, except that the upper temperature shall be 52.7° ± 3.5° Celsius instead of 71° Celsius. Within 1 hour after completion of the test, the power supply shall be examined for corrosion or other physical damage and shall be subjected to the performance test procedures of 4.7.2.

4.7.3.8.1 Humidity, operating. If specified by the procuring activity, the power supply shall be subjected to the performance tests of 4.7.2, while at 52.7° ± 3.5° Celsius and 85 percent relative humidity during the last cycle of the humidity test of 4.7.3.8.

4.7.3.9 Salt fog. The power supply shall be tested in accordance with MIL-STD-810, method 509, procedure I. After the test, the power supply shall be examined for evidence of corrosion which affects operation. After drying and removal of salt deposits, the power supply shall be subjected to the procedure of 4.7.1.

4.7.3.10 Fungus. If specified by the procuring activity, the power supply shall be tested in accordance with MIL-STD-810, method 508, procedure I. After the test, the power supply shall be examined for evidence of deterioration, corrosion, or fungus growth, and the power supply shall be subjected to the procedure of 4.7.1.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the power supply shall be preserved in accordance with MIL-P-116, method II, and packaged and packed in accordance with MIL-STD-794.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Shipment nomenclature for marking shall be as a minimum "POWER SUPPLY 2639691."

6. NOTES

6.1 Intended use. The power supply is intended for use in Test Set Mark 281 Mod 1. It provides all internal direct-current power, 33.5 volts alternating current for internal use, and torpedo warm-up direct-current power.

6.2 Ordering data. Procurement documents should specify, but not be restricted to the following:

(a) Title, number, and date of this specification

(b) Power supply drawing number: NAVORD Drawing 2639691

(c) The quanity ordered should include test units that will be expended as a result of first article inspection

(d) Information concerning inclusion of optional environmental tests and/or exceptions to this or other pertinent documents

(e) Level of packaging and packing desired, if other than as specified in 5.1.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project Number 6130-N075)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-P-82579 (OS), Power Supply, Multivoltage, Direct Current (for Torpedo Test Set Mark 281 Mod 1)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

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